

CHAPTER 26: TRAFFIC SIGNAL INSTALLATION

26.1 INTRODUCTION

- 1 The installation of traffic signals requires a significant amount of planning and design by skilled and knowledgeable professionals. A high degree of attention to detail is required. Often, the cost of planning and design is high, compared with the cost of the traffic signal installation itself, but the benefits far exceed such cost.
- 2 Warrants for the installation of traffic signals are given in Chapter 2 of this manual. The installation is warranted when the traffic signals can a) meet all the *minimum requirements* as described in this manual, b) no viable and feasible *alternative solution* is available which, when implemented, would obviate the need for traffic signals and c) the traffic signals meet the minimum *queue length warrants*.
- 3 The investigation of signal sites and installation of traffic signals require the following tasks:
 - (a) Candidate site identification.
 - (b) Warrant study.
 - (c) Signal design.
 - (d) Signal installation.
 - (e) Commissioning.The road authority accepts responsibility for traffic signal operations during the commissioning phase of the project.
- 4 Checklists are given in Appendix C to this manual that can be used for checking signal designs and for approving of traffic signals. The checklists should be signed by the responsible registered professional engineer or technologist of the road authority.

26.2 CANDIDATE SITE IDENTIFICATION

- 1 Candidate locations for the installation of traffic signals can be identified by means of a variety of methods. Many locations are identified following requests from the public or decision-makers. Traffic engineers and technologists in the employment of a road authority can also contribute in this regard.
- 2 The queue length warrant used for justifying the installation of traffic signals is a simple method of identifying possible locations for signalisation. Observations over a short period of time during peak hours at a junction or a pedestrian crossing would indicate the presence of long queues of vehicles.
- 3 A site should initially be inspected visually to establish whether it is likely that the installation of traffic signals would not be warranted. During this inspection, attention should be given to the minimum requirements for the installation of signals.

26.3 WARRANT STUDY

- 1 Once a candidate site for signalisation has been identified, a study should be undertaken to establish whether the installation of traffic signals would be warranted according to the requirements of Chapter 2 of this manual.
- 2 The study must start off by establishing whether a traffic signal can be installed without violating the minimum requirements for traffic signal installations as set out in this manual. The installation of traffic signals would not be warranted when any of the minimum requirements cannot be met.
- 3 The next step in the warrant study is to establish whether no viable and feasible alternative solution other than traffic signals is available which, when implemented, would obviate the need for traffic signals. Traffic signals would not be warranted if such a solution is available.
- 4 The final step in the study is to undertake a queue length study to establish whether the queue length warrants given in Chapter 2 will be met. A traffic signal installation would be warranted if the site passes this final test.
- 5 When a traffic signal is warranted, the site can be placed on a priority list until funds become available for the installation of the signal. Priority should be given to those locations with the longest queues (as measured during the warrant study).

26.4 SIGNAL DESIGN

- 1 Once a traffic signal has been warranted at a junction or crossing, the design of the signal can proceed. Traffic studies should be undertaken, the site must be surveyed and the traffic signal designed. Contract documents and specifications will also be required if a contractor undertakes the work.
- 2 A proper land survey should be made of the site showing all relevant information such as:
 - (a) Property boundaries and fences.
 - (b) Carriageways, kerbs, shoulders, islands, medians, existing road markings.
 - (c) Paved sidewalks, driveways.
 - (d) Drainage structures.
 - (e) Plants and vegetation (including location, size and spread of larger trees).
 - (f) Engineering services (telephone, electricity, water, sanitation, etc).
 - (g) Roadside furniture such as bus shelters, telephone booths, retaining walls, guard rails and light poles.
 - (h) Adjacent buildings together with any awnings, height above the kerb and distance back from the kerb face.
 - (i) Any other structures such as bridges, retaining walls, fills and cuts, etc.

- 3 It is important that attention should be given to the possible geometric improvement of a junction during the design phase. Particular attention should be given to the provision of auxiliary turning lanes, particularly right-turn lanes, but also possibly left-turn lanes. Auxiliary straight-through lanes may also be required.
- 4 The site should be regularly visited and inspected during the design stage to ensure that the plan will be satisfactory. Checks should be made during such visits to ensure that the survey included all relevant information. **It is the responsibility of the design engineer or technologist to ensure that the survey was undertaken correctly.**
- 5 During the site visits, attention should be given to possible problems such as:
 - (a) Inadequate space for traffic signal placement.
 - (b) Available sight distances on all approaches and for all turning movements.
 - (c) Distance to adjacent traffic signal sites.
 - (d) Location of any nearby emergency services that may require priority.
 - (e) Most appropriate location for the controller.
 - (f) Condition of road pavement for installation of loop detectors.
 - (g) Source of supply of power.
 - (h) Parking spaces for signal maintenance vehicles.
- 6 It would be beneficial if a qualified electrician could also inspect the site to identify any possible electrical problems that may occur. The actual source and availability of power must be confirmed.
- 7 Proposed designs should be discussed with all parties with an interest in the design to ensure agreement. This includes the electronic and electrical engineers and other departments of the road authority. Where other authorities are also affected, it is particularly important that they should form part of the process from the outset of the project.
- 8 Design plans should contain full information on all details related to the traffic signal. Fewer problems can be expected during installation if sufficient care has been taken during the design phase. Designs should also take future maintenance needs into account. It may be possible to introduce a number of relatively simple design features that could reduce maintenance costs substantially.
- 9 The following design plans would normally be required:
 - (a) Junction or crossing design showing the geometric design, road signs and markings.
 - (b) Traffic signal layout plan showing the location of traffic signal faces, signal posts, overhead gantries of cantilevers, loop detectors and the controller.
 - (c) Duct diagram, indicating the position of ducts and draw boxes.
 - (d) Existing engineering services plan, indicating which services have to be relocated.
 - (e) Traffic signal timing and phasing diagrams.

26.5 SIGNAL INSTALLATION

- 1 Successful traffic signal installation depends on effective supervision and control during installation. A high degree of supervision is required to ensure that the signal is installed according to specifications.
- 2 Installation can either be done by the road authority or by private contractors. Larger road authorities that install traffic signals regularly would probably install signals departmentally. Smaller road authorities would need to appoint private contractors for such work.
- 3 The utilisation of qualified and experienced contractors can significantly reduce the possibility of installation faults. The possibility of future liability resulting from such faults should be an important consideration when selecting contractors. Preference should be given to contractors with the required experience, skills, knowledge and resources to complete a project.
- 4 A suitably qualified person should be assigned by the road authority to be responsible for direct supervision of signal installation contracts. Electrical work should be inspected and approved by a qualified electrician.
- 5 Before commencing with installation, the contractor should submit a detailed timetable or schedule in advance for approval by the project engineer or technologist. The typical installation sequence for traffic signal installations is:
 - (a) Civil engineering work
 - (b) Underground.
 - (i) Footings.
 - (ii) Ducts and draw boxes.
 - (iii) Cables, earthing and wiring.
 - (iv) Detector loops.
 - (c) Above ground.
 - (i) Signal posts and overhead installations.
 - (ii) Traffic signal heads.
 - (iii) Electrical wiring and conduits.
 - (iv) Cabinet and control equipment.
 - (v) Electrical connection.
 - (d) Testing of installed signals.
- 6 Particular attention must also be given to the **accommodation of traffic** during the installation. Traffic signal faces should be masked, and existing controls maintained at the junction. The possibility of liability claims resulting from accidents should always be a prime concern.
- 7 During the contract, adequate records must be kept of the progress with the installation of the signals. Any delays must be carefully documented and reported. A daily diary should be kept in which all events and relevant information are logged. The information can be summarised in a number of status reports.
- 8 Any changes to the initial design should be properly authorised and logged (using variation orders). Changes may cover such items such as increased or decreased quantities, design alterations, equipment substitution, etc.

26.6 COMMISSIONING

- 1 Before a traffic signal is finally commissioned, it is imperative that the installation be properly checked and inspected, and traffic signal operations reviewed.
- 2 During this check, all signal plans should be tested and light signals inspected for proper operation. Once the proper operation of traffic signal equipment has been ascertained, traffic operations should be tested for a short period of time for possible serious traffic conflicts. A traffic officer should be present during the testing period.
- 3 A commissioning checklist is provided in Appendix C to assist with the commissioning of a signal. The checklist should be used as a supplement to, and not as a replacement for, the contract specifications and specific instructions of product manufacturers and suppliers. The checklist should be signed by the responsible registered professional engineer or technologist of the road authority.
- 4 It is important that the traffic signal should be monitored after it has been brought into operation. Traffic patterns may change as a result of the signal installation, while drivers may not respond to signals as anticipated.
- 5 Changed traffic patterns will require updating of traffic signal timings once traffic patterns have stabilised. At least one month should be allowed for traffic to stabilise. Traffic may then have to be recounted and new traffic signal timings and phasing determined. The provision of turning phases may even require modifications to the traffic signal face layout.

